

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI - K K BIRLA GOA CAMPUS

Semester-I (2025/2026)

Analog & Digital VLSI Design (EEE/INSTR F313)

Quiz-I (PART B)

Date : 09.09.2025

Duration : 30 Minutes

Maximum Marks : 15

Name of the student:

Palak Kshetrapal

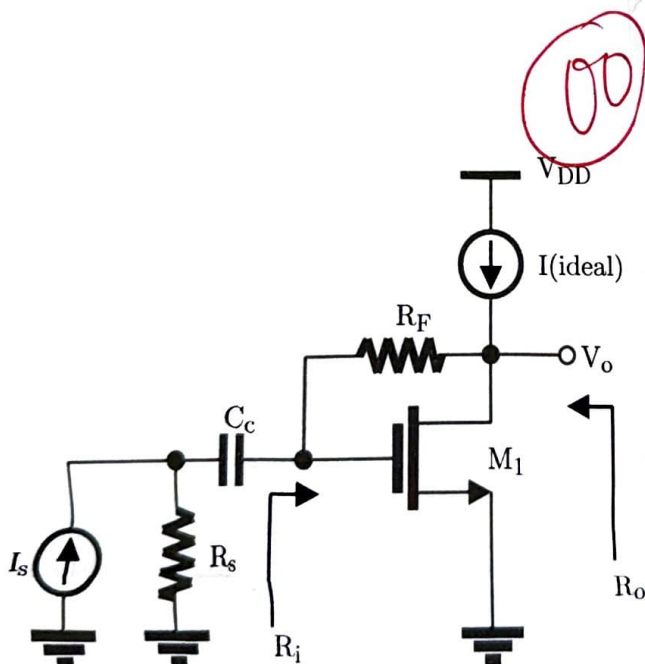
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Question	Marks	Score
1	15	00
Total:	15	00

Question 1

Given $g_{m1}=2 \text{ mA/V}$, $r_{o1}=40 \text{ k}\Omega$, $R_F=20 \text{ k}\Omega$, $R_S=2 \text{ k}\Omega$, $C_c=\infty$, find the parameter values specified in the table. Answers without proper unit and/or sign will be marked invalid. [15]



A	80 V/V
β	0.909 0.0909
$A\beta$	7.2727 V/V
A_f	9.6704 V/V
R_i	400 Ω
R_o	4400 Ω
$R_{i,f}$	30.839 3309.08 Ω
$R_{o,f}$	531.8699 Ω